Cart lab: change mass

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a controlled comparison to describe the effect of system mass on acceleration.

SUCCESS CRITERIA - I can...

1. Identify total moving mass and keep driving force fixed.
2. Compare acceleration values using a fair test.
3. Separate model data from measured evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “mass” and “system” to explain your model or result. Sentence starters are optional.

mass: a measure of how strongly an object resists acceleration

system: the object or objects chosen for analysis

acceleration: rate of change of velocity

01 Bring back an earlier idea

A car starts at +24 m/s and brakes at -4 m/s² until rest. A student extends that acceleration to 8 s.
Find the stop time/distance and explain why extension is invalid for ordinary braking.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $a = F_{net}/M = m_h\ g/M$.
Ideal cart + hanger: M is all moving mass; m_h is hanging mass. Hold m_h fixed; neglect resistance and pulley inertia.

Worked reasoning: Supplied cart trials use the same net force: masses 1 and 2 kg accelerate at 4 and 2 m/s². In both cases ma=4 N; doubling mass halves acceleration.

Your turn: Plot acceleration against mass for the two trials.
Label what was held constant.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Test an inverse relationship

The ideal data use the same 0.49 N net force. Compute 1/M for each row and describe the graph of a versus 1/M.
Give slope units and physical meaning.

Word help: ideal = using stated simplifying conditions

Sentence starter: The ... in my model represents ... . Its label shows ... .

M (kg)	a (m/s²)
0.50	0.980
0.75	0.653
1.00	0.490
1.50	0.327

Marzano check - circle: ① ② ③

04 System boundary

Explain why the hanger belongs in total moving mass but string tension is internal to the cart-plus-hanger system.
State two idealizations.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Measured slope and repetitions

One supplied v-t interval goes from (0.20 s, 0.10 m/s) to(0.80 s, 0.40 m/s).
Find a.
Design a three-mass, three-trial data table for the actual lab; label any measurements you collect.

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

06 Choose a method

A 3 kg object hangs still from a vertical string.
Use g=9.8 N/kg.
Name all forces on it with sources, sizes and directions. Does object-on-string belong on this diagram?

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: 1 2 3

07 Exit check

A 0.80 kg moving system accelerates at 0.60 m/s². At the same net force, predict a for 1.20 kg.
Explain how added resistance would change the prediction.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A cart model predicts a=F/m. At constant net force 3 N, find accelerations for masses 0.5, 1, 2 kg.
Choose a transformed horizontal axis that makes a straight-line graph and state its slope.

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3